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PRODUCT MANUAL

产品手册

FAIRINO

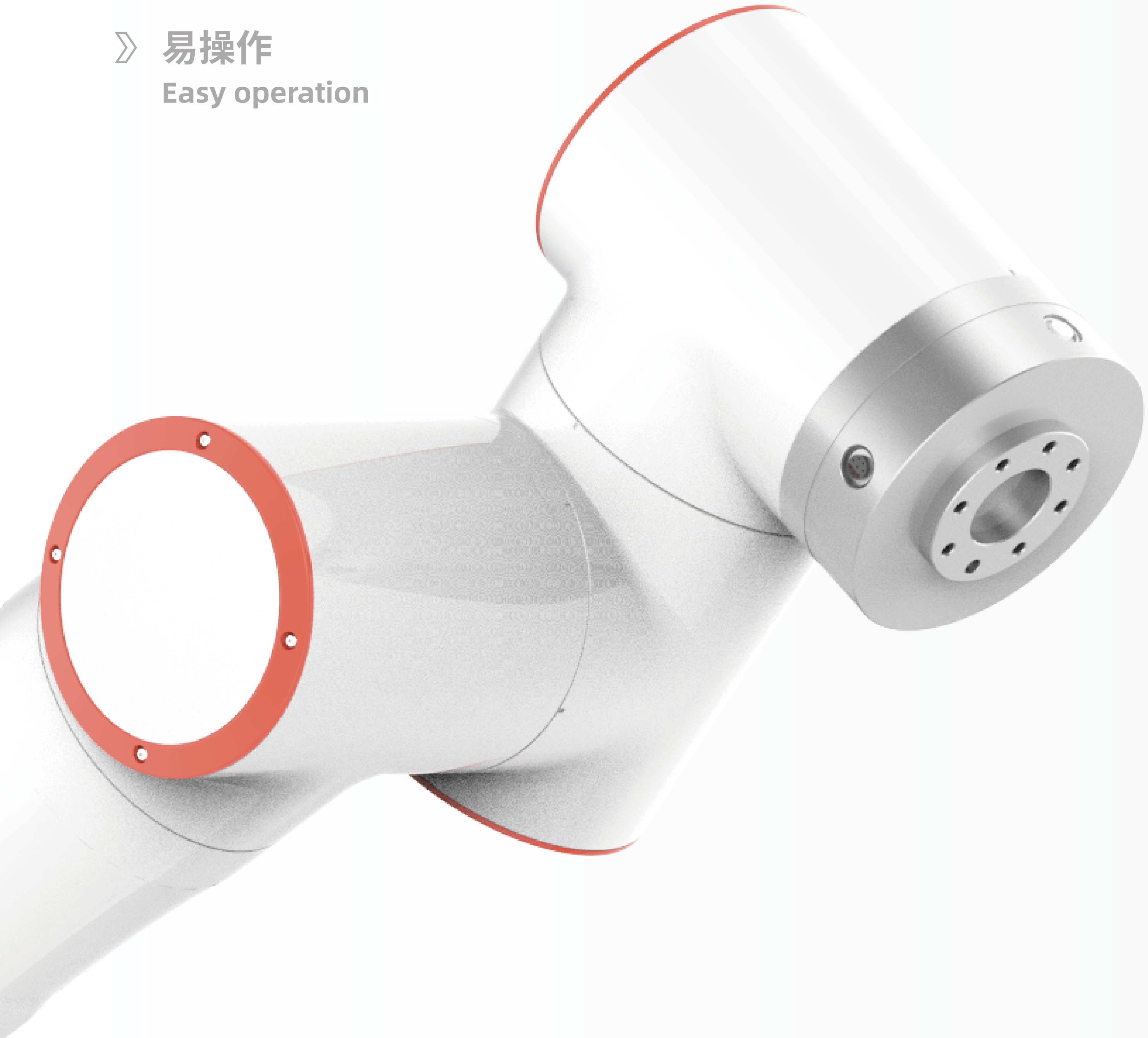
法奥意威(苏州)机器人系统有限公司
FAIR Innovation (Suzhou) Robot System Co.,Ltd.

》模块化
Modularization

》快部署
Quick deployment

》易操作
Easy operation

FAIRINO
ROBOT



PRODUCT VISION
产品愿景

协作机器人将延伸您的时间和空间，解放繁杂的低效重复，让您的双手和大脑拥抱更广袤的天际。未来，您将见证**法奥协作机器人**与众协同。
Collaborative robots will extend your time and space, liberate complex and inefficient repetition, and let you embrace a wider horizon.In the future, You will see **FAIRINO ROBOTS**.

随着它的出现，不仅提高了人机协作的效率，也加快了更多企业自动化进程，对许多制造商来说，它还节省了空间，降低了部署机器人产线的成本。
With its emergence, it not only improves the efficiency of human-machine collaboration, but also speeds up the automation process for more enterprises and frees floor space and lowers the cost of implementing robots for manufacturers.

FAIRINO

PRODUCT DISPLAY
产品展示

FAIRINO FR SERIES

系列排布

根据不同的负载能力和参数范围，法奥协作机器人产品FR系列拥有七个型号：FR3MT、FR3（可选镜像版本）、FR5、FR10、FR16、FR20和FR30。法奥通过国际权威机构已获取更全产品认证，为伙伴以及客户提供更优品质保障。

According to different payload and parameter, FAIRINO collaborative robots FR series are divided into seven models:FR3MT, FR3(Optional Mirror Version), FR5, FR10, FR16, FR20 and FR30. To provide partners&customers with better quality assurance,FAIRINO has obtained a more comprehensive range of certificates through international certification organizations.

法奥协作机器人FR系列产品生产通过了ISO 9001质量管理体系认证

Quality Management System: ISO 9001

产品认证：CR, CE, KCs, NRTL, RoHS 2.0, NSF, SEMI, IP65

Product Certification: CR, CE, KCs, NRTL, RoHS 2.0, NSF, SEMI, IP65

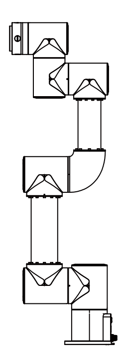
ISO功能安全认证：ISO 10218, ISO 13849, ISO 15066

ISO Functional Safety Certification: ISO10218, ISO13849, ISO15066

智能人机系统解决方案
Intelligent human-robot cooperation system solutions

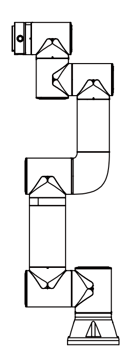
FR3MT

有效负载 Payload: 3kg(瞬时5kg)
工作半径 Reach: 622mm
重复定位精度 Repeatability: ±0.05mm



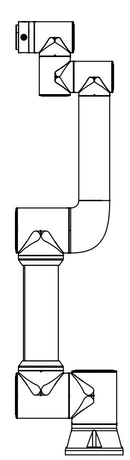
FR3

有效负载 Payload: 3kg
工作半径 Reach: 622mm
重复定位精度 Repeatability: ±0.02mm



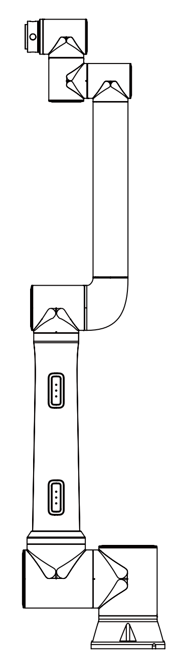
FR5

有效负载 Payload: 5kg
工作半径 Reach: 922mm
重复定位精度 Repeatability: ±0.02mm



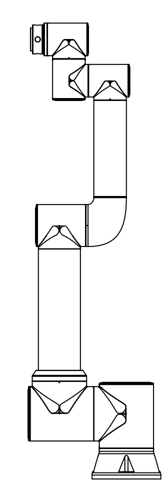
FR10

有效负载 Payload: 10kg
工作半径 Reach: 1400mm
重复定位精度 Repeatability: ±0.05mm



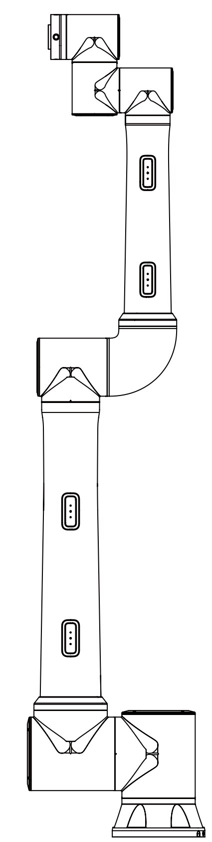
FR16

有效负载 Payload: 16kg
工作半径 Reach: 1034mm
重复定位精度 Repeatability: ±0.03mm



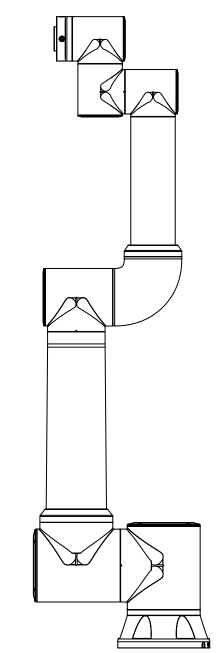
FR20

有效负载 Payload: 20kg
工作半径 Reach: 1854mm
重复定位精度 Repeatability: ±0.1mm



FR30

有效负载 Payload: 30kg
工作半径 Reach: 1403mm
重复定位精度 Repeatability: ±0.1mm



ROBOT ARM TECHNICAL SPECIFICATION

机械臂规格参数

	FR3MT		FR3		FR5		FR10		FR16		FR20		FR30	
有效负载 (Payload)	**3kg (瞬时5kg)**		**3kg**		**5kg**		**10kg**		**16kg**		**20kg**		**30kg**	
工作半径 (Reach)	**622mm**		**622mm**		**922mm**		**1400mm**		**1034mm**		**1854 mm**		**1403 mm**	
自由度 (Degrees of freedom)	6个旋转关节		6个旋转关节		6个旋转关节		6个旋转关节							
人机交互 (HMI)	10.1 英寸示教器或移动终端 Web App 10.1 inch teach pendant or mobile terminal Web App								10.1 英寸示教器或移动终端 Web App 10.1 inch teach pendant or mobile terminal Web App					
双臂机器人应用 (Dual arm robotics applications)			有镜像版本，可组建双臂机器人											
符合ISO 9283的位姿可重复性 (Pose repeatability per ISO 9283)	**±0.05mm**		**±0.02mm**		**±0.02mm**		**±0.05mm**		**±0.03mm**		**±0.1mm**		**±0.1mm**	
轴移动(Axis movement)	工作范围	最大速度	工作范围	最大速度	工作范围	最大速度	工作范围	最大速度	工作范围	最大速度	工作范围	最大速度	工作范围	最大速度
基座 (Base)	±175°	±150°/s	±175°	±180°/s	±175°	±180°/s	±175°	±120°/s	±175°	±120°/s	±175°	±120°/s	±175°	±120°/s
肩部 (Shoulder)	+ 85°/ – 265°	±150°/s	+ 85°/ – 265° (双臂: – 85°/ + 265°)	±180°/s	+ 85°/ – 265°	±180°/s	+ 85°/ – 265°	±120°/s	+ 85°/ – 265°	±120°/s	+ 85°/ – 265°	±120°/s	+ 85°/ – 265°	±120°/s
肘部 (Elbow)	±150°	±150°/s	±150°	±180°/s	±160°	±180°/s	±160°	±180°/s	±160°	±180°/s	±160°	±120°/s	±160°	±120°/s
腕部 1 (Wrist 1)	+ 85°/ – 265°	±180°/s	+ 85°/ – 265° (双臂: – 85°/ + 265°)	±180°/s	+ 85°/ – 265°	±180°/s	+ 85°/ – 265°	±180°/s	+ 85°/ – 265°	±180°/s	+ 85°/ – 265°	±180°/s	+ 85°/ – 265°	±180°/s
腕部 2 (Wrist 2)	350°	±180°/s	±175°	±180°/s	±175°	±180°/s	±175°	±180°/s	±175°	±180°/s	±175°	±180°/s	±175°	±180°/s
腕部 3 (Wrist 3)	±175°	±180°/s	±175°	±180°/s	±175°	±180°/s	±175°	±180°/s	±175°	±180°/s	±175°	±180°/s	±175°	±180°/s
典型 TCP 速度(Typical TCP speed)	1m/s		1m/s		1m/s		1.5m/s		1m/s		2m/s		2m/s	
防护等级(IP classification)	IP54 (可选 IP65)		IP54 (可选 IP65)		IP54 (可选 IP65)		IP54 (可选 IP65)		IP54 (可选 IP65)		IP54 (可选 IP65)		IP54 (可选 IP65)	
噪音(Noise)	<65dB		<65dB		<65dB		<65dB		<65dB		<70dB		<70dB	
安装方向(Robot mounting)	任何方向		任何方向		任何方向		任何方向		任何方向		任何方向		任何方向	
I/O端口(I/O Ports)	数字输入(DI) 2	数字输出(DO) 2	数字输入(DI) 2	数字输出(DO) 2	数字输入(DI) 2	数字输出(DO) 2	数字输入(DI) 2	数字输出(DO) 2	数字输入(DI) 2	数字输出(DO) 2	数字输入(DI) 2	数字输出(DO) 2	数字输入(DI) 2	数字输出(DO) 2
	模拟输入(AI) 1	模拟输出(AO) 1	模拟输入(AI) 1	模拟输出(AO) 1	模拟输入(AI) 1	模拟输出(AO) 1	模拟输入(AI) 1	模拟输出(AO) 1	模拟输入(AI) 1	模拟输出(AO) 1	模拟输入(AI) 1	模拟输出(AO) 1	模拟输入(AI) 1	模拟输出(AO) 1
工具I/O电源 (Tool I/O power supply)	24V/1.5A		24V/1.5A		24V/1.5A		24V/1.5A		24V/1.5A		24V/1.5A		24V/1.5A	
底座直径(Footprint)	125mm		128mm		149mm		190mm		190mm		240mm		240mm	
整机重量(Weight)	≈10kg		≈15kg		≈22kg		≈40kg		≈40kg		≈85kg		≈85kg	
工作温度(Operating temperature)	0-45℃		0-45℃		0-45℃		0-45℃		0-45℃		0-45℃		0-45℃	
工作湿度(Operating humidity)	90%RH(non-condensing)		90%RH(non-condensing)		90%RH(non-condensing)		90%RH(non-condensing)		90%RH(non-condensing)		90%RH(non-condensing)		90%RH(non-condensing)	
设备材料(Materials)	铝、钢		铝、钢		铝、钢		铝、钢		铝、钢		铝、钢		铝、钢	
黑色选配（Black Optional)	否(no)		否		是		是		否		否		否	

■ 典型功率测试负载设置，根据机器型号设置不同的负载,负载配置参数如下：
Typical power test payload settings, different loads are set according to robot models, payload configuration parameters are as follows :

	FR3MT	FR3	FR5	FR10	FR16	FR20	FR30
	FR3MT负载设置:3kg，Z轴:18mm FR3MT payload setting: 3kg, Z-axis: 18mm	FR3负载设置:3kg，Z轴:18mm FR3 payload setting: 3kg, Z-axis: 18mm	FR5负载设置:5kg，Z轴:30mm FR5 payload setting: 5kg, Z-axis: 30mm	FR10负载设置:10kg，Z轴:60mm FR10 payload setting: 10kg, Z-axis: 60mm	FR16负载设置:16kg，Z轴:96mm FR16 payload setting: 16kg, Z-axis: 96mm	FR20负载设置:20kg，Z轴:120mm FR20 payload setting: 20kg, Z-axis: 120mm	FR30负载设置:30kg，Z轴:200mm FR30 payload setting: 30kg, Z-axis: 200mm

选用老化测试程序，机器人总电源接入功率计，机器人设置自动模式，全局速度设置为100，点击运行,运行两个周期无异常则开始持续测试，持续24小时。24小时后分别统计功率计的峰值和平均功率，依次对各个型号进行统计：
Select aging test program, connect robot's total power to power meter, set robot to automatic mode, set global speed to 100, click run, if there are no abnormalities after running two cycles, start continuous testing for 24 hours. After 24 hours, respectively, record the peak and average power of the power meter, and then statistically analyze each model :

	FR3MT	FR3	FR5	FR10	FR16	FR20	FR30
典型平均功率 (Typical average power)	200W	220W	260W	300W	310W	620W	600W
典型峰值功率 (Typical peak power)	230W	280W	310W	500W	410W	810W	910W

CONTROLLER TECHNICAL SPECIFICATION
控制箱规格参数




	直流mini控制箱2kW	直流控制箱5kW	交流mini控制箱2kW	交流控制箱5kW
设备特性 Features				
防护等级(IP classification)	IP54	IP54	IP54	IP54
工作温度(Operating temperature)	0-45℃	0-45℃	0-45℃	0- 45℃
工作湿度(Operating humidity)	90%RH(non-condensing)	90%RH(non-condensing)	90%RH(non-condensing)	90%RH(non-condensing)
I/O端口(I/O Ports)	数字输入(DI) 16　数字输出(DO) 16	数字输入(DI) 16　数字输出(DO) 16	数字输入(DI) 16　数字输出(DO) 16	数字输入(DI) 16　数字输出(DO) 16
	模拟输入(AI) 2　模拟输出(AO) 2	模拟输入(AI) 2　模拟输出(AO) 2	模拟输入(AI) 2　模拟输出(AO) 2	模拟输入(AI) 2　模拟输出(AO) 2
	高速脉冲输入(High speed pulse input) 2	高速脉冲输入(High speed pulse input) 2	高速脉冲输入(High speed pulse input) 2	高速脉冲输入(High speed pulse input) 2
I/O电源(I/O power supply)	24V/1.5A	24V/1.5A	24V/1.5A	24V/1.5A
标配通讯(Standard communication)	I/O、TCP/IP、Modbus_TCP/RTU	I/O、TCP/IP、Modbus_TCP/RTU	I/O、TCP/IP、Modbus_TCP/RTU	I/O、TCP/IP、Modbus_TCP/RTU
可选通讯(Optional communication)	CC-Link、Profinet、Ethernet/IP、EtherCAT	CC-Link、Profinet、Ethernet/IP、EtherCAT	CC-Link、Profinet、Ethernet/IP、EtherCAT	CC-Link、Profinet、Ethernet/IP、EtherCAT
通讯板卡选配 (Communication Board Optional Configuration)	MiniPCI Express- 实时以太网 PC板卡	MiniPCI Express- 实时以太网 PC板卡	MiniPCI Express- 实时以太网 PC板卡	MiniPCI Express- 实时以太网 PC板卡
软件开发包 (Software development kit)	C#/C++/Python/ROS/ROS2	C#/C++/Python/ROS/ROS2	C#/C++/Python/ROS/ROS2	C#/C++/Python/ROS/ROS2
物理性能 Physical				
尺寸参数(L*W*H)	245*180*44.5mm(不含凸出物)	245*180*89mm(不含凸出物)	245*180*44.5mm (不含凸出物)	245 *180*89mm (不含凸出物)
设备重量(Weight)	2.1kg (不含线重量)	2.957 kg (不含线重量)	2.5kg (不含线重量)	3.6 kg (不含线重量)
设备材料(Materials)	镀锌板　Galvanized plate	镀锌板　Galvanized plate	镀锌板　Galvanized plate	镀锌板　Galvanized plate
供电电源(Power supply)	30-60VDC	30-60VDC	176-264VAC ~ 50-60Hz 100-240VAC ~ 50-60Hz	100-240VAC ~ 50-60Hz
适配机器人 Applicable Robot	FR3,FR5,FR10,FR16	FR20/FR30	FR3,FR5,FR10,FR16	FR20/FR30

SAFETY BOX
按钮盒

防护等级(IP classification)	IP54
按钮功能(Button function)	手动/自动、拖动、点记录、是否配合按钮盒、开始/停止、关机 Manual/Auto, Drag, Point Record, Match or Not with Safety Button Box, Start/Stop, Shutdown
协议类型(Communication)	TCP/IP
网络传输速率(Network transfer rate)	100M
供电(Power over ethernet)	标准POE Standard POE
尺寸参数(L*W*H)	136*60*66mm (不含凸出物) (No protrusions)
设备重量(Weight)	490g (带线重量) (Cable weight included)
设备材料(Materials)	ABS
线缆长度(Cable length)	5m
按键次数(Number of keys)	≥20W 次

人机交互工具,可实现基础交互功能。通过RJ45接口可与电脑、平板等设备链接,直接登录到web示教界面。

Human-cobot interaction tools for basic interaction functions. It can be linked with computers, tablets and other devices through the RJ45 interface, and directly log in to the Web App teaching interface.

TEACH PENDANT [选配 Optional]
示教器

防护等级(IP classification)	IP54
工作湿度(Operating humidity)	90%RH(non-condensing)
显示分辨率(Display resolution)	1280 x 800 pixels
尺寸参数(L*W*H)	268*210*88mm
设备重量(Weight)	1.6kg
设备材料(Materials)	ABS、PP
线缆长度(Cable length)	5m

示教器、电脑、平板或手机与WebAPP系统连接,从而实现对协作机器人的操控。

The teach pendant, computer, tablet or mobile phone is connected to the WebAPP system to realize the operation of the collabroative robot.

EXPLOSION-PROOF CABINET [选配 Optional]
防爆柜

防爆柜(整体防爆)
Explosion-Proof Cabinets (Integral Explosion-Proof)

机器人(全系列防爆)
Robot (full range explosion-proof)

具有良好的技术性能。机柜应具有抗振动、抗冲击、耐腐蚀、防尘、防水、防辐射等性能,以便保证设备稳定可靠地工作。良好的使用性和安全防护设施,便于操作、安装和维修,并能保证操作者安全。

It has good technical performance. The cabinet must be resistant to vibration, impact, corrosion, dust, water, and radiation to ensure stable and reliable operation of devices. Good usability and safety protection facilities, easy operation, installation and maintenance, and to ensure the safety of the operator.

防护等级(IP classification)	IP65
工作环境温度(Operating Environment Temperature)	0°C-45°C
最低正压(Minimum Positive Pressure)	100pa
最高正压(Maximum Positive Pressure)	1000pa
额定输入电压(Rated Input Voltage)	KW/48VDC/A
最大泄漏量(Maximum Leakage)	15L/Min
尺寸参数(L*W*H)	682*500*1100(含报警灯1286)
设备重量(Weight)	75kg
设备材料(Materials)	碳钢+不锈钢
柜内容积(Cabinet Volume)	126L
换气流量(Ventilation Flow Rate)	120L/min
换气时间(Ventilation Time)	14min
保护气体(Protective Gas)	空气(AIR)
防爆柜证书(Explosion-proof Cabinet Certificate)	CE22.7131

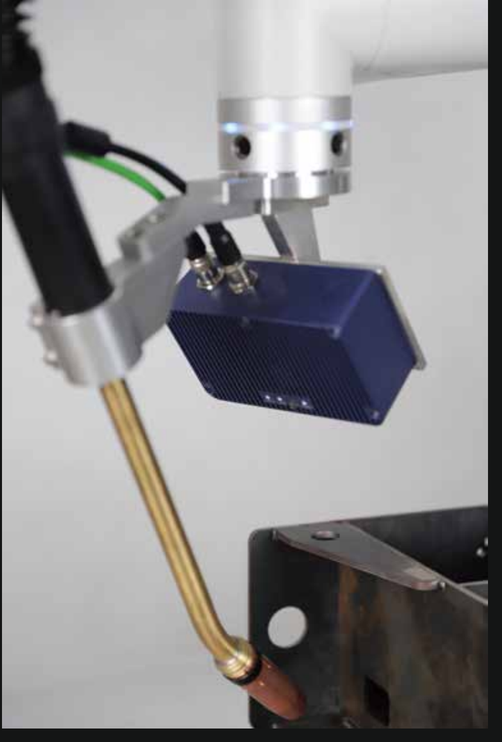
INDUSTRY APPLICATIONS

上下料解决方案

Pick And Place Solution

上下料机器人能够提高生产效率、质量和安全性，降低劳动强度，并提供灵活适应性，为企业带来更高的效益和竞争优势。

Material handling robots can improve production efficiency, quality, and safety, reduce labor intensity, and provide flexibility and adaptability, bringing higher benefits and competitive advantages to businesses.



Welding Robot

焊接机器人

丰富的焊接工艺包，具备多种焊接工艺，包含有点焊、段焊、直焊、摇摆焊、圆弧焊、多层多道焊，并且具备焊丝寻位、焊缝追踪的智能焊接技术，显著提升焊接效率，保障焊接品质。

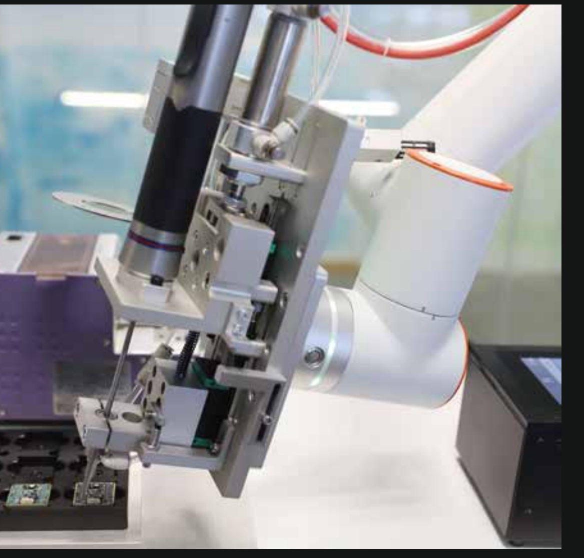
Abundant welding process kits, with a variety of welding technologies such as spot welding, seam welding, straight welding, oscillating welding, arc welding, and multi-layer multi-pass welding. It also incorporates intelligent welding technologies for wire positioning and weld seam tracking, significantly enhancing welding efficiency and ensuring welding quality.

Screw Tightening Robot

螺丝锁付机器人

搭配末端智能拧紧装置，实现了扭矩可调、可控、可编辑，适合各种场景下的螺丝锁付，能够稳定、高效、准确完成生产过程，极大减少工人重复性劳动，支持数据溯源。

Combined with the end intelligent tightening device, it achieves adjustable, controllable, and programmable torque, making it suitable for screw locking in various scenarios. It can stably, efficiently, and accurately complete the production process, greatly reducing repetitive labor for workers and supporting data traceability.



涂胶解决方案

Glue Dispending Solution

搭配末端智能出胶装置，精细化作业，适合多场景下的精细化涂胶、点胶工作，能够稳定、高效、准确完成涂胶过程，保证涂胶效果，极大减少工人重复性劳动，保护工人健康。

Paired with an intelligent dispensing device at the end effector, it enables precise operations and is suitable for precise gluing and dispensing tasks in various scenarios. It can achieve stable, efficient, and accurate adhesive application, ensuring the quality of the adhesive work. This greatly reduces repetitive labor for workers and protects their health.

传送带解决方案

Conveyor Belt Solution

- 提升工作安全性 Enhance work safety
- 实时监控与反馈 Real-time monitoring and feedback
- 降低错误率和损耗 Reduce error rate and losses
- 提高生产效率 Improve production efficiency
- 数据记录与溯源 Data recording and traceability
- 精确追踪与识别 Accurate tracking and identification

COMMERCIAL APPLICATIONS

茶饮机器人
Automated Tea Robot

节约人力成本，替代人工，提高工作效率；冲泡茶饮口感一致，改善不同人操作、不同时间点操作而带来的差异性；具有表演性，给消费者带来乐趣，而员工可以去完成更有成就感、待遇更高的工作；成本低，快速回本，经济效益好；占地面积小，坪效更高，能适应多种新颖的创业模式。

协作机器人可以应用在多类型的新零售场景，并且根据不同场景需求，进行个性化定制。

Collaborative robots can be applied in various types of new retail scenarios and can be customized according to different scenario requirements. Benefits include:

Cost-saving: They replace manual labor, reducing manpower costs while increasing work efficiency.

Consistent tea brewing: They ensure consistent taste regardless of different operators or different time points, eliminating variations caused by human factors.

Entertainment value: The robotic performance brings enjoyment to consumers, while employees can focus on more fulfilling and higher-paying jobs.

Cost-effective: They have low costs and provide a quick return on investment, resulting in good economic benefits.

Small footprint: They occupy less space, resulting in higher space utilization and adaptability to various innovative business models.

康复解决方案
Rehabilitation Solution

实现了上肢康复与下肢锻炼的一体化，通过运动轨迹的复现，降低了使用门槛。通过实时记录反馈数据，显著提升安全性能。多种模式设定，让康复治疗更有针对性，显著提升康复效率。

It has achieved integration of upper limb rehabilitation and lower limb exercise, reducing the barrier to entry through the reproduction of motion trajectories. By recording real-time feedback data, it significantly enhances safety performance. With various mode settings, it makes rehabilitation treatment more targeted, leading to a significant improvement in rehabilitation efficiency.

艾灸解决方案
Moxibustion Solution

完整复刻五大灸法，提供悬停灸、雀啄灸、回旋灸、往复灸和循经灸，降低艾灸门槛。经过最新认证，搭配末端碰撞检测、温控、红外测距，三重防护保障艾灸安全。内置吸入装置，避免艾灸过程烟尘吸入。

- 极致安全
- 部署灵活
- 门槛降低
- 高效艾灸

It fully replicates the five major moxibustion techniques, offering hovering moxibustion, sparrow pecking moxibustion, rotating moxibustion, reciprocating moxibustion, and meridian moxibustion, thus reducing the barrier to entry for moxibustion. With the latest certifications, it is equipped with end collision detection, temperature control, and infrared distance measurement, providing triple protection to ensure the safety of moxibustion.

It also has a built-in suction device to prevent inhalation of smoke and dust during the moxibustion process.

- Ultimate safety
- Flexible deployment
- Lower barrier to entry
- Efficient moxibustion

COMPANY PROFILE
公司简介

FAIRINO
ROBOT

FAIRINO，优先实现全核心零部件自主研发的协作机器人公司。

我们关注用户体验，致力于为行业提供便利的深度智能系统解决方案。

我们为行业客户提供定制化的部件、整机及系统，开放的开发平台为我们的合作伙伴提供更多的便捷和可能。

与伙伴和客户共同高速成长，为伙伴和客户提供超额价值。

欢迎来到FAIRINO的智能世界。

FAIRINO is the collaborative robot company who has achieved independent R&D of all core components.

We focus on user experience and are dedicated to offering the industry with artificial intelligent robot system.

We provide customized components, complete machines and systems for industry customers, the open development platform provides more convenience and possibility for our partners.

FAIRINO, as always, provides values and grow together with customers and partners.

Welcome to the intelligent world of FAIRINO.

随着智能制造时代的到来，许多行业迈入"智能"行列，协作机器人能为其做什么呢？

协作机器人为企业用户降本增效，提升员工技能；为商业用户提供更优质服务，优化客户使用体验；为手工业者制定标准化流程，降低制作成本；为家庭成员分担家务，辅助清洁，烹饪工作。

相信协作机器人具有无限潜力，未来可以应用到更多的场景里。

Lots of manufacturers have begun taking advantage of AIoT and human-robot collaboration. What can collaborative robots do for them?

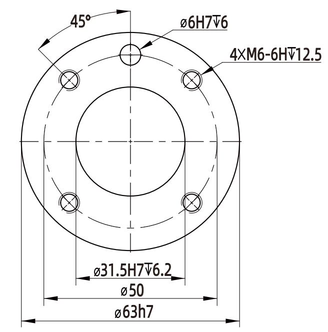
Collaborative robots decrease manufacturing costs, increase the efficiency of production and enhance the skills of employees. They also offer better service quality and improve the customer experience. By providing the standardized functions and low deploying costs, cobots are widespread in commercial scenarios such as household chores, room cleaning and cooking.

Cobots are believed to have unlimited potential and would be introduced to more scenarios in the future.

DRAWINGS

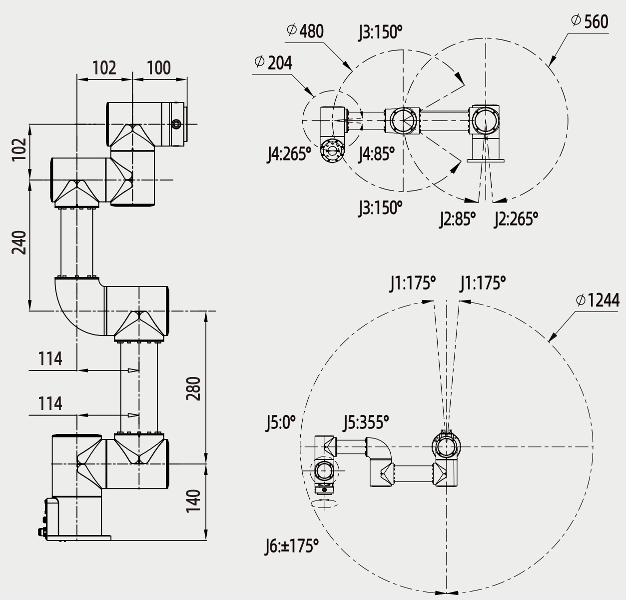
技术图纸

单位(Unit) : mm

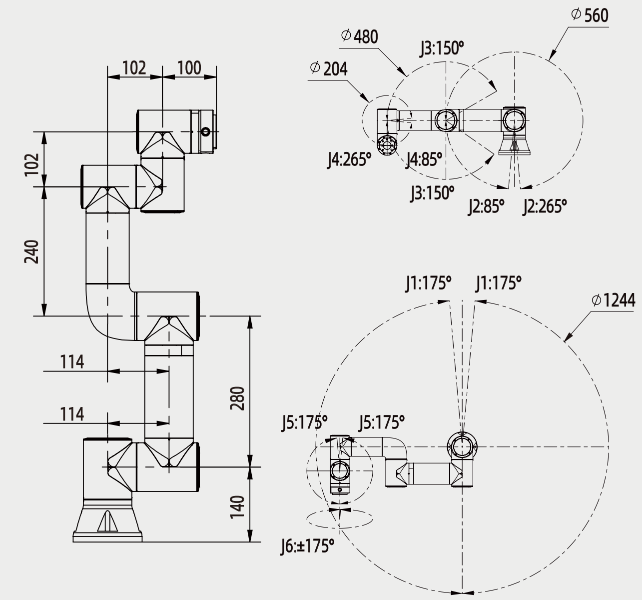


机器人末端
均采用国际标准

ROBOT END-EFFECTOR COMPATIBLE WITH
INDUSTRIAL ROBOT END-EFFECTOR CONNECTION METHODS



FR3MT



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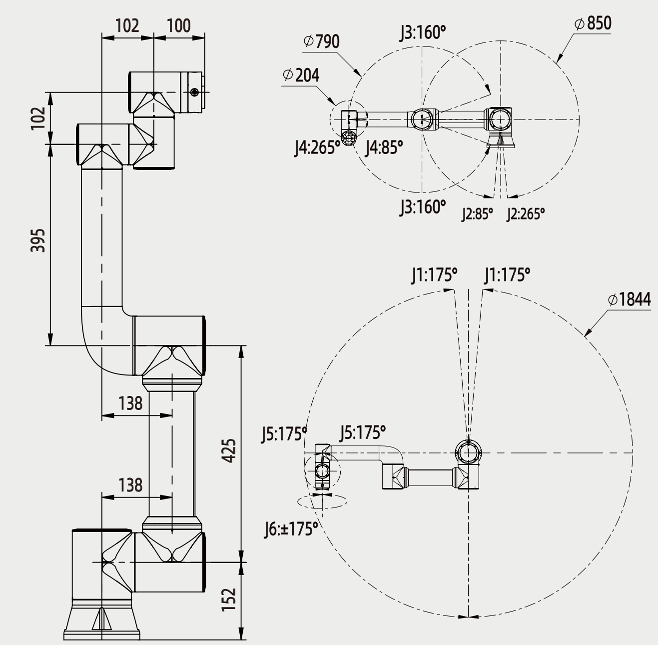

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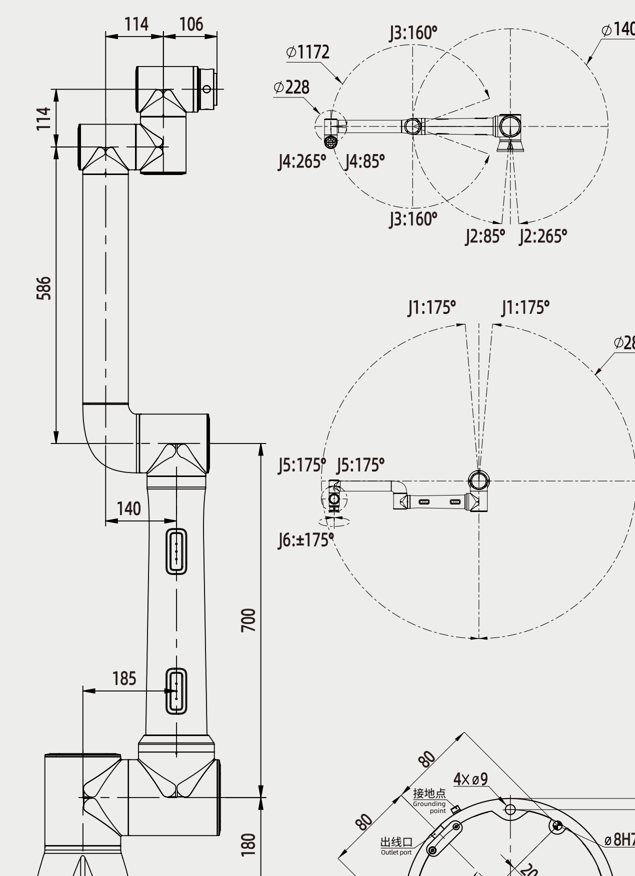

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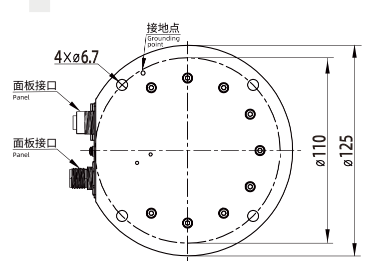
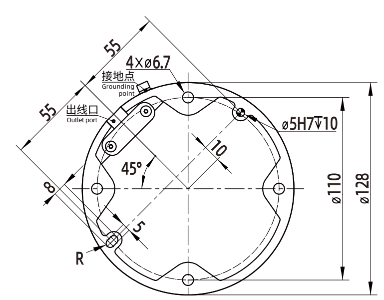
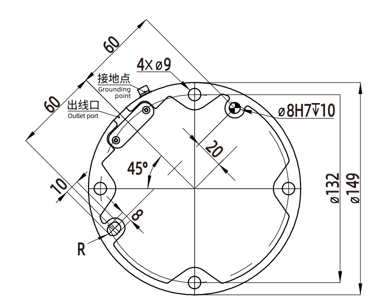


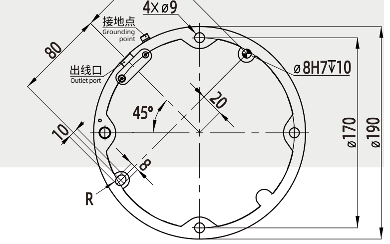
FR3MT 基座图
Pedestal diagram



FR3 基座图
Pedestal diagram



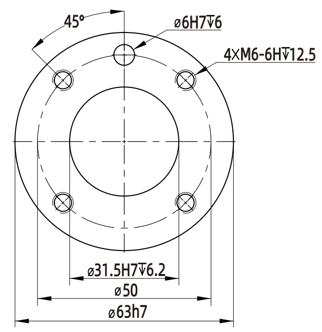
FR5 基座图
Pedestal diagram



FR10 基座图
Pedestal diagram

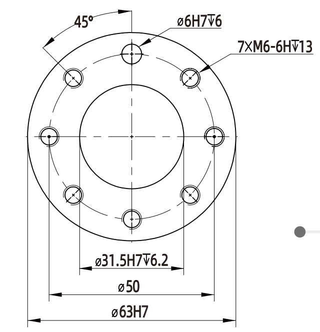
DRAWINGS
技术图纸

单位(Unit) : mm



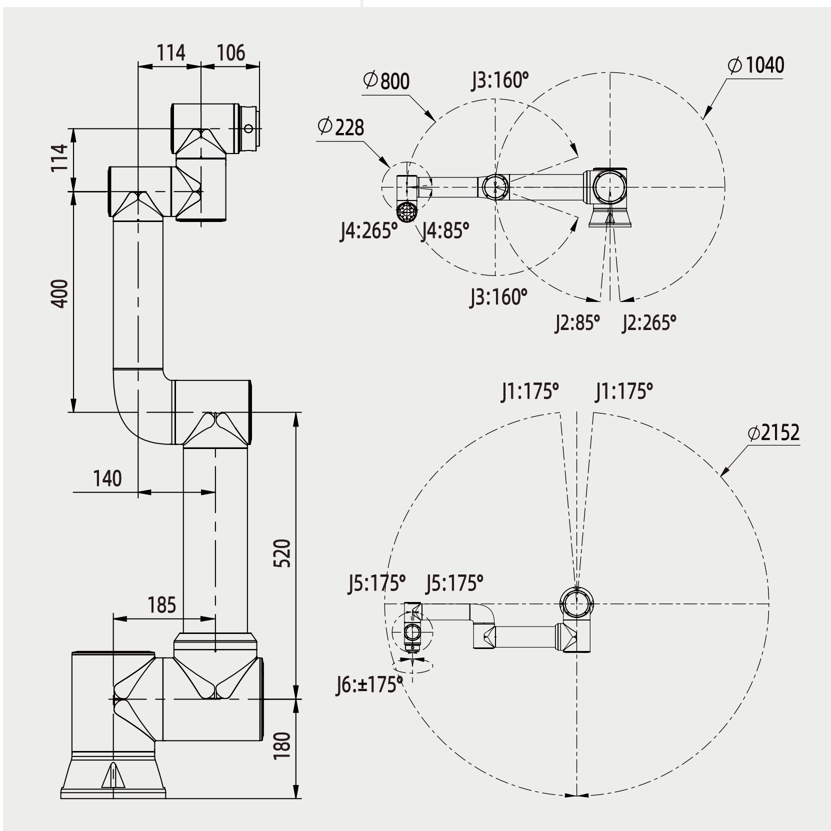
机器人末端
均采用国际标准

ROBOT END-EFFECTOR COMPATIBLE WITH
INDUSTRIAL ROBOT END-EFFECTOR CONNECTION METHODS



机器人末端兼容
工业机器人末端连接方式

ROBOT END-EFFECTOR COMPATIBLE
WITH INDUSTRIAL ROBOT END-EFFECTOR CONNECTION METHODS



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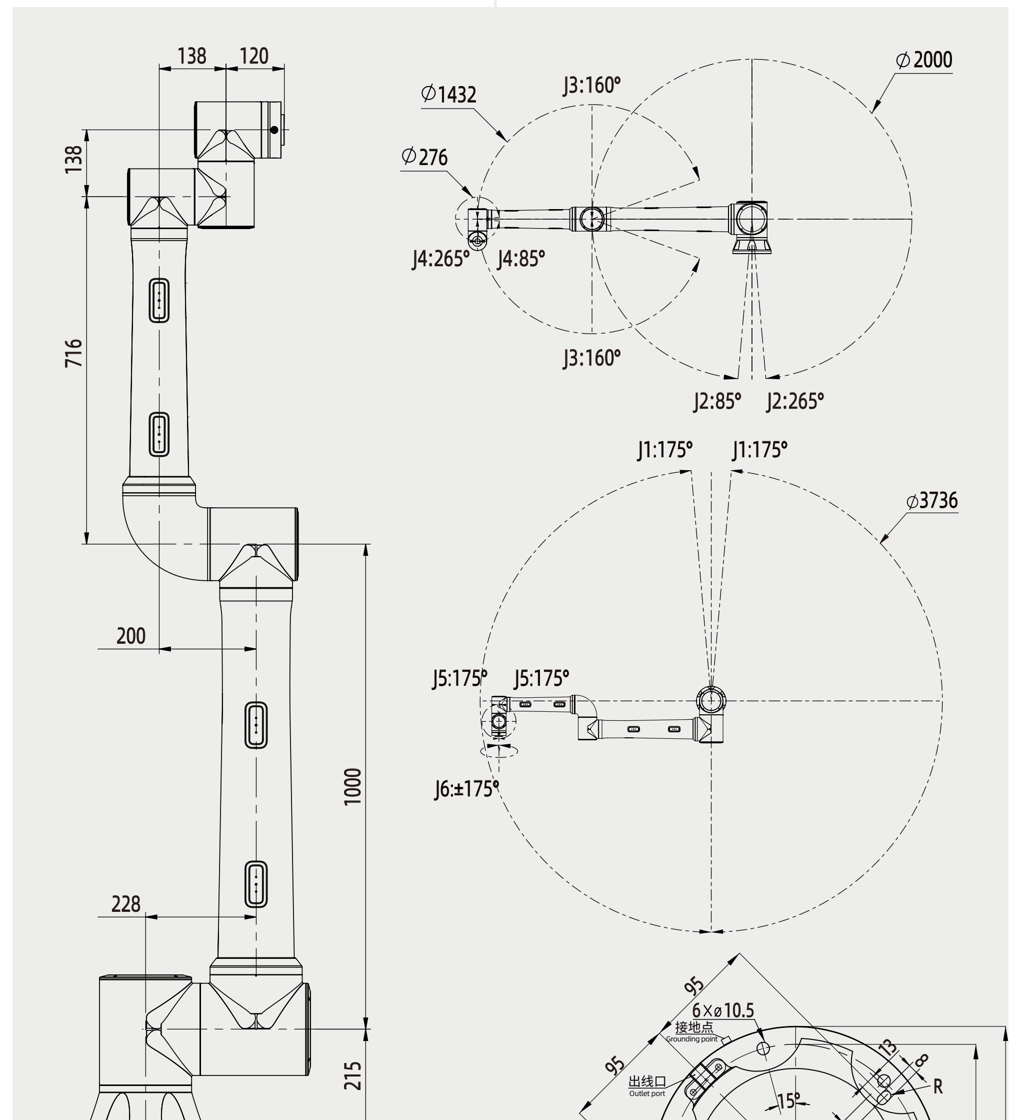

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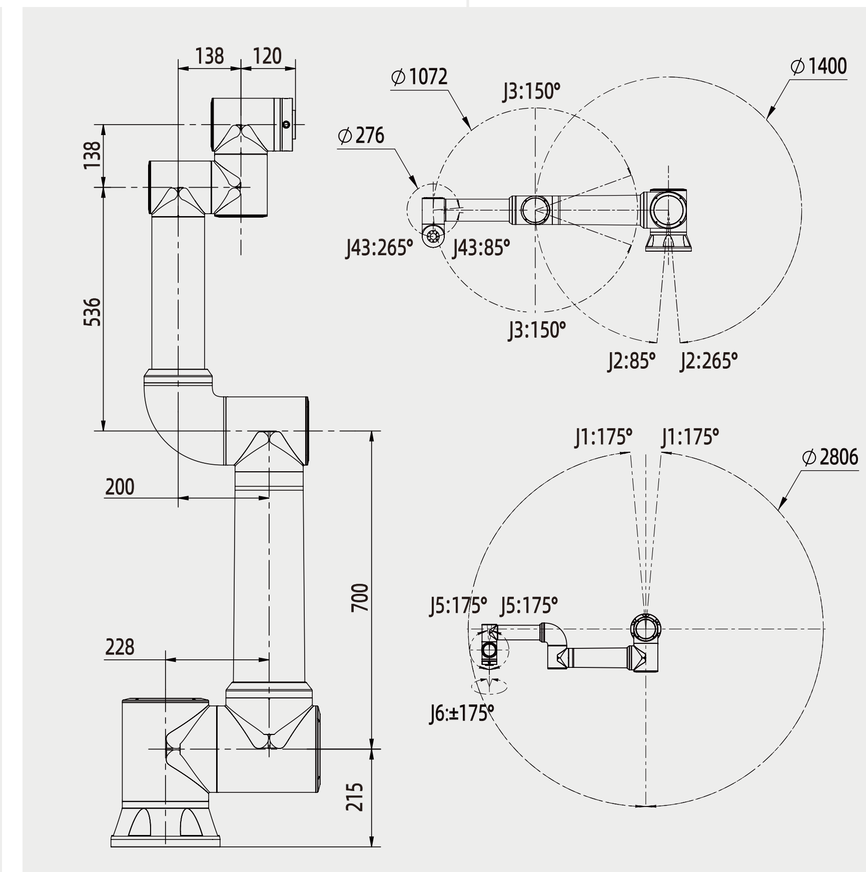
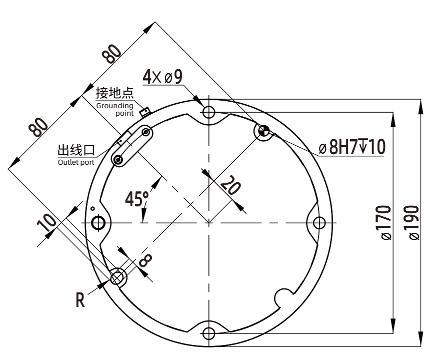
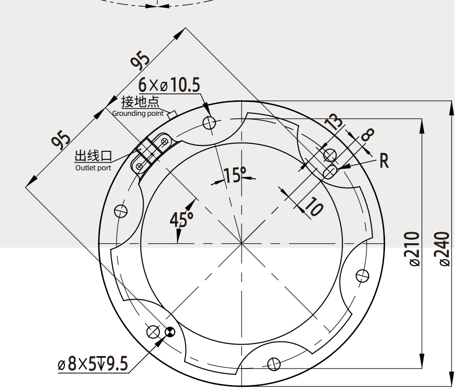


FR30



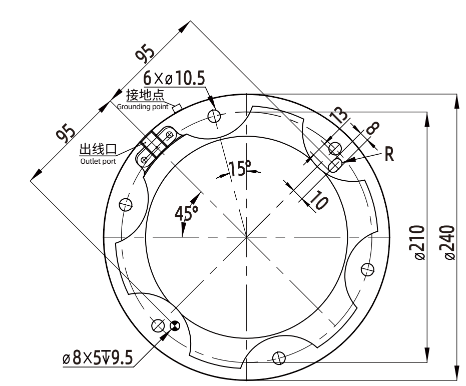
FR16 基座图

Pedestal diagram



FR20 基座图

Pedestal diagram



FR30 基座图

Pedestal diagram